



[RISK ASSESSMENT REPORT: UAV WAVE PROJECT]

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EXECUTIVE SUMMARY

UAV wave systems present multiple cyber-physical risks that could affect service continuity, operational control, and public safety if deployed at city level. The highest priority risk identified is disruption to the dispatch control server, as this function coordinates UAV routing, scheduling, and operational allocation across the infrastructure. Its compromise would have immediate, system-wide consequences and could prevent the safe management of UAV services. Administrative system compromise is also a major issue, as privileged access could allow attackers to alter configurations, disable infrastructure, or impact command functions. UAV navigation manipulation is a higher priority risk because modified positioning data could directly affect route integrity and flight safety.

Additionally, medium to high risks include UAV communication interference, compromise of antenna infrastructure, degradation of the Security Operations Centre monitoring, and manipulation of operational data. While these risks may not always cause a complete service breakdown, they can affect flexibility, delay incident detection, and weaken confidence in the system to function properly.

Overall, the analysis shows that centralised control systems, communication infrastructure, and navigation mechanisms need the highest security level because disruption in these areas would have the greatest operational effect. The key suggestions are to strengthen administrative access controls through multi-factor authentication and strict privilege separation, improve resilience of dispatch systems through redundancy and network segmentation, secure UAV communications and navigation signals through encryption and integrity protection, and enhance SOC monitoring through continuous logging and anomaly detection. These measures should be considered before wider deployment because they significantly reduce the likelihood and impact of the most severe risks.

RISK ASSESSMENT METHODOLOGY:

NIST SP 800-30 Revision 1 was chosen for this evaluation because the UAV Wave system is a complex cyber-physical environment that combines distributed communication infrastructure, centralised control functions, privileged administrative access, real-time monitoring, and safety-critical UAV operations. The methodology is appropriate for this situation because it supports consistent assessment of threat sources, vulnerabilities, likelihood, impact, and overall risk across interconnected digital and physical assets.

Its application compares risks affecting the dispatch control server, the administrative control system, the communication network, the antenna

infrastructure, the monitoring capability, and the UAV navigation functions. This is vital in a city-scale mobility system, where interference to a single control component can affect multiple services all at once.

A key advantage of this approach is that it supports clear and structured prioritisation of risks across different assets and threat events. This makes it suitable for assessing infrastructure where both operational continuity and public safety are important. A limitation is that likelihood and impact values rely on analyst-defined scoring criteria, which can introduce partiality. To reduce this, consistent scoring levels were applied to the dataset using a risk-severity model explained later in the report.

THREAT MODELLING EXERCISE

Attack Tree analysis was used to illustrate how attackers could affect the UAV Wave system. This approach fits the scenario because the infrastructure relies on connected UAVs, communication networks, control servers, administrative interfaces, SOC monitoring, and operational data.

The top event is loss of safe operational command and control of the UAV Wave system. In this setting, that would disrupt routing, communication, and monitoring, reduce service reliability and increase safety risk. The main attack paths identified are administrative system compromise, dispatch control server disruption, UAV communication interference, antenna infrastructure compromise, SOC monitoring failure, UAV navigation manipulation, and operational data manipulation. These paths form the basis of the cyber-physical security events analysed later in the report.

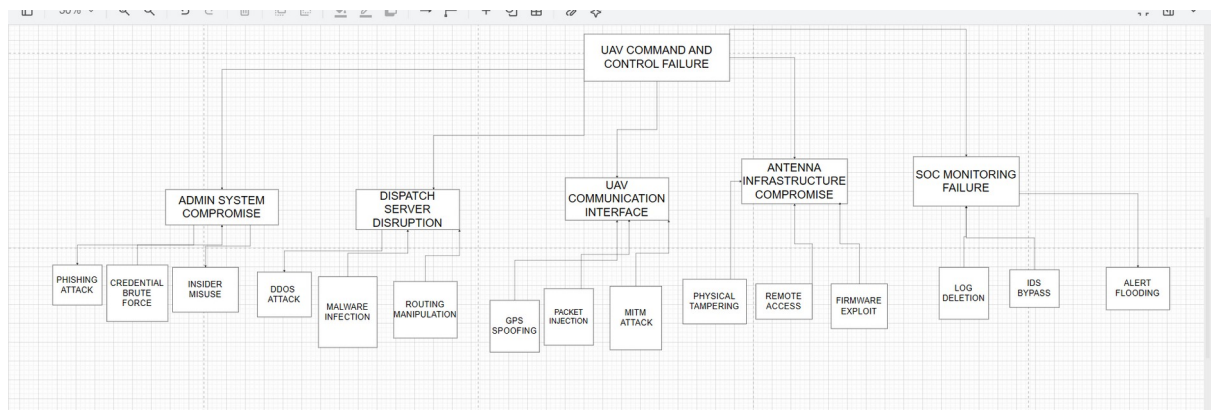


Fig 1: Attack Tree for loss of safe operational command and control of the UAV wave system

The cyber-physical security events analysed in this report were obtained directly from the main attack tree shown in Figure 1

ASSET IDENTIFICATION

Asset identification was used to establish which parts of the UAV Wave infrastructure need a high security level. In this evaluation, assets are the system components that allow UAV mobility services to operate safely, reliably, and at scale.

The UAV Wave system links physical UAV equipment with communication networks, control systems, monitoring platforms, and operational data services. Compromise of these assets may affect both the performance of the digital system and physical UAV operations. This is vital in a city-scale environment where service continuity, control reliability, and public safety are closely related.

The identified assets support the threat modelling exercise and the later risk analysis. They represent the main operational components influencing system availability, control, communication, and safe service delivery.

ASSET IDENTIFICATION TABLE

Asset ID	Asset Name	Description	Importance of the System
A1	UAV Fleet	Autonomous aerial vehicles used for transportation and delivery services	Core operational components enabling UAV mobility services
A2	Communication Network	Wireless infrastructure connecting UAVs, antennas and control systems	Allows real-time command, telemetry and coordination
A3	Dispatch Control Server	Central servers are responsible for UAV routing, scheduling, and operational coordination.	Ensures safe navigation and coordination of UAV operations
A4	Administrative control system	Privileged management interface used by permitted operators and administrators	Provides Configuration and operational control of the system
A5	Antenna Infrastructure	Distributed antenna nodes are deployed across neighbourhoods	Maintains reliable communication between UAVs and control centres

		for network connectivity.	
A6	System Monitoring System	Operational dashboards and monitoring systems are used to detect incidents and anomalies.	Enables security monitoring and incident response
A7	Operational Data and Telemetry	Flight data, sensor readings and system status information generated by UAVs	Vital for navigation safety and system monitoring

Cyber-Physical Security Events

The UAV Wave infrastructure functions as a cyber-physical system where digital control platforms directly communicate with physical UAV operations, communication networks, and monitoring services. Due to this connection, security weaknesses in cyber components can affect UAV routing, service provision, and safety. To enable data driven evaluation, a dataset was analysed in Microsoft Excel, featuring simulated UAV activities, SOC alerts, access logs, antenna events, and dispatch records. During risk analysis, patterns within these records assist in spotting recurring operational anomalies, unauthorised privileged access, communication failures, monitoring problems, and navigation irregularities. These patterns were grouped into seven security events, based on the main attack tree branches and key system assets identified earlier. The result illustrates primary threat scenarios that threaten the reliability, control, communication, monitoring, and safety of the UAV Wave system.

Event 1: Administrative System Compromise

The administrative control system grants access to system configuration, user management, and operational settings. If an attacker gains access through phishing, password guessing, or authentication weaknesses, they could alter core functions, withdraw access, disable infrastructure, or interfere with routing. Because the administrative system interfaces with multiple operational components, a compromise could cause widespread disruption. Likelihood is moderate, but impact is high due to the level of control provided.

Event 2: Dispatch Control Server Disruption

The dispatch control server coordinates UAV assignments, routing, and resource allocation. Disruption through malware, vulnerabilities, or denial-of-service attacks could prevent UAVs from receiving instructions or updates. In a city-wide network, this may quickly affect deliveries, emergency services, and wider operations. Because the dispatch server is significant to UAV mobility, both likelihood and impact are high.

Event 3: UAV Communication Interference

UAV operations rely on continuous communication between aerial vehicles, antennas, and control centres. Attackers may disrupt this by jamming, intercepting, or injecting malicious packets. Communication failure may delay control signals, interrupt telemetry, or reduce operator awareness, creating safety and operational risks. Likelihood is moderate, while impact remains high because communication is essential for safe operation.

Event 4: Antenna Infrastructure Compromise

Antenna infrastructure supports communication between UAVs and control systems. If attackers interfere with or abuse antenna equipment, they could reduce coverage, interrupt data transmission, or create blind spots. Because several UAV operations may rely on the same antenna node, a compromise could affect multiple services at once. The likelihood is lower than for purely digital events, but the impact is still significant.

Event 5: SOC Monitoring System Failure

The SOC provides real-time monitoring and incident detection. Attackers may attempt to bypass it by modifying logs, disabling sensors, or flooding dashboards. If SOC capability is degraded, ongoing attacks may remain hidden, allowing attackers to persist. The likelihood is low to moderate, but the impact is high because safe system operation relies on effective monitoring.

Event 6: UAV Navigation Manipulation

Navigation manipulation impacts UAV flight behaviour by altering GPS or positioning data. Attackers may spoof signals, leading to route deviations, inaccuracies, or entry into restricted zones. In urban environments, this poses safety hazards and can disrupt deliveries, passenger transport, or emergency services. The likelihood is moderate, but the impact is high due to the direct physical consequences.

Event 7: Operational Data Manipulation

Operational data includes telemetry, sensor readings, maintenance records, and infrastructure reports. If this data is altered, operators may make wrong decisions based on false information. This may delay response, affect routing, or hide faults within the system. Likelihood is low to moderate, but impact remains significant because operational decision-making depends on accurate data.

Risk Analysis

The identified cyber-physical security events were analysed using a qualitative risk assessment approach supported by the dataset generated for this study. The dataset was created and processed in Microsoft Excel to simulate UAV operations, SOC alerts, administrative access logs, antenna activity, and dispatch events. It was used not only to identify the main events, but also to support likelihood, impact, and risk scoring.

Likelihood shows the probability that a threat source could successfully exploit a vulnerability within the UAV Wave infrastructure. Likelihood scores were based on the relative frequency and repetition of simulations in the dataset, including repeated privileged-access failures, dispatch anomalies, communication issues, and monitoring alerts. Assets with wider exposure and more frequent incident patterns were assigned higher likelihood values.

Impact shows the severity of consequences if a threat event successfully affects the system. Impact scores were based on the level of disruption observed in the dataset, including service degradation, loss of control, reduced monitoring visibility, communication failures, and navigation errors. In a UAV environment, these consequences affect both service continuity and operational safety.

Risk severity was calculated using the formula:

$$\text{Risk} = \text{Likelihood} \times \text{Impact}$$

Likelihood and impact were scored on a 1 to 5 scale. Risk levels were classified as Low (1-5), Medium (6-10), High (11-15), and Critical (16-25). For example, dispatch control server disruption was assigned a likelihood score of 4 and an impact score of 5, producing a risk score of 20, which was classified as Critical. Events affecting centralised control and routing functions produced the highest risk values because they could disrupt multiple UAV services simultaneously.

Risk Evaluation and Matrix

A 5×5 risk matrix was used to analyse and prioritise the identified events, with likelihood and impact scores ranging from 1 to 5. Risk levels were classified as Low (1-5), Medium (6-10), High (11-15), and Critical (16-25). The treated dataset was used to compare final scores and position each event within the matrix.

The matrix showed dispatch control server disruption as the highest risk. Administrative system compromise and UAV navigation manipulation were also high risks. Communication interference and compromised antenna infrastructure posed medium-to-high risks, while SOC monitoring failure and operational data manipulation posed medium risks. This evaluation was used to identify which events required immediate treatment and which required stronger monitoring and resilience measures.

*Table 1: 5*5 risk matrix used for likelihood-impact prioritisation:*

Impact/ Likelihood	1	2	3	4	5
5	5	10	15	20	25
4	4	8	12	16	20
3	3	6	9	12	15
2	2	4	6	8	10
1	1	2	3	4	5

Recommended Security Controls

This focuses on reducing the highest rated risks identified during the assessment. For the administrative control system, strong authentication, strict privilege management, and ongoing monitoring of privileged accounts are important, as a breach could allow attackers to alter configurations and disrupt operations. Regarding dispatch control servers, redundancy, secure configuration, and network segmentation are advised since these central coordination points are critical; their

failure could impact multiple UAV operations simultaneously, so resilience measures are vital to prevent service interruptions. For UAV and antenna communications, encrypted and authenticated protocols are recommended to reduce the risk of interception, spoofing, and signal manipulation. Additionally, improved logging, enhanced anomaly detection, and continuous SOC monitoring are necessary, as delayed detection could allow attacks to persist and intensify. To protect antenna infrastructure, physical safeguards and tamper detection are recommended, given its distributed nature and susceptibility to interference. These controls were prioritised because they target the most significant risks and enhance system resilience, safety, and operational continuity.

REFERENCES AND APPENDIX

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